

Data Analytics Project 1 – Data Cleaning and Preparation

Submitted By:

Yanamala Manisha

Internship:

DecodeLabs Data Analytics Internship

Date:

June 2026

Objective:

The objective of this project was to clean, validate, and prepare an e-commerce dataset for further analysis by identifying missing values, duplicate records, date format issues, and calculation inconsistencies.

Dataset Overview:

- Dataset Type: E-commerce Transactions
- Total Records: 1200 - Total Columns: 14

Data Cleaning Steps

1. Missing Value Analysis

The dataset was examined for missing values across all columns. CouponCode values were checked and no missing values remained after validation.

2. Duplicate Record Validation

Duplicate OrderIDs were identified using SQL GROUP BY and HAVING clauses. No duplicate OrderIDs were found in the dataset.

3. Date Format Validation

The Date column was validated against the YYYY-MM-DD format. All records contained valid dates and no invalid date entries were found.

4. Data Type Verification

Data types were reviewed and confirmed for numerical and text fields including Quantity, UnitPrice, TotalPrice, CustomerID, Product, and TrackingNumber.

5. Price Consistency Validation

TotalPrice values were compared with $\text{Quantity} \times \text{UnitPrice}$. Minor differences were identified due to decimal rounding. No significant pricing inconsistencies were found.

Results

- Total Records: 1200
- Missing Values: 0
- Duplicate OrderIDs: 0
- Invalid Dates: 0
- Significant Price Errors: 0

Conclusion:

The dataset was successfully cleaned and validated. No duplicate records or invalid dates were identified. Missing values were handled appropriately, and price calculations were verified. The dataset is now suitable for further exploratory data analysis and business insights generation.

SCREENSHOTS:

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: project x project

SCHEMAS

Filter objects

▼ decode labs_project1

- Tables
 - dataset for data analytic
 - ecommerce_data
- Views
- Stored Procedures
- Functions

▼ parks_and_recreation

- Tables
 - employee_demographic
 - employee_salary
 - parks_departments
- Views
- Stored Procedures
- Functions

▼ sys

Administration Schemas

Information

Schema: decode labs_project1

Object Info Session

Limit to 1000 rows

```
2 • create database decode labs_project1;
3 • use decode labs_project1;
4
5 • select *
6   from ecommerce_data;
7
8 -- TOTAL RECORDS COUNT
```

Result Grid

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	No Coupon Applied	Email	268.56
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42

ecommerce_data 27 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:58:53	select * from ecommerce_data LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.015 sec

Read Only

33°C Mostly sunny

Search

ENG IN

11:02 18-06-2026

fig.1 ECOMMERCE DATA

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: project

SCHEMAS

Filter objects

▼ decode labs_project1

Tables

dataset for data analytic

ecommerce_data

Views

Stored Procedures

Functions

▼ parks_and_recreation

Tables

employee_demographic

employee_salary

parks_departments

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

Schema: decode labs_project1

Object Info Session

Limit to 1000 rows

```
6 from ecommerce_data;
7
8 -- TOTAL RECORDS COUNT
9 select COUNT(*) AS Total_Records
10 from ecommerce_data;
11
12
```

Result Grid

Total_Records
1200

Filter Rows: Export: Wrap Cell Content: 12

Result 28

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	10:58:53	select * from ecommerce_data LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.015 sec
✓ 2	11:03:37	select COUNT(*) AS Total_Records from ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec

Light rain In the afternoon

Search

ENG IN

11:03 18-06-2026

fig 2: TOTAL RECORDS COUNT

mysql workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

project x project

Limit to 1000 rows

12 -- MISSING VALUE VALIDATION
13 • SELECT COUNT(*) AS missingCouponCode
14 From ecommerce_data
15 WHERE couponcode IS NULL
16 or couponcode= ''
17
18
19

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

missingCouponCode
0

Administration Schemas

Information

Schema: decodealabs_project1

Object Info Session

Result 29 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	10:58:53	select * from ecommerce_data LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.015 sec
✓ 2	11:03:37	select COUNT(*) AS Total_Records from ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
✓ 3	11:06:25	SELECT COUNT(*) AS missingCouponCode From ecommerce_data WHERE couponcode IS NULL or couponc...	1 row(s) returned	0.000 sec / 0.000 sec

33°C Mostly sunny

Search

11:06 18-06-2026

fig 3: MISSING VALUE VALIDATION

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: project x project

SCHEMAS Open a SQL script file in a new query tab

Filter objects

- ▼ decode labs_project1
 - Tables
 - dataset for data analytic
 - ecommerce_data
 - Views
 - Stored Procedures
 - Functions
- ▼ parks_and_recreation
 - Tables
 - employee_demographic
 - employee_salary
 - parks_departments
 - Views
 - Stored Procedures
 - Functions
- ▼ sys

Administration Schemas

Information

Schema: decode labs_project1

ecommerce_data 32 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 4	11:12:35	SELECT *, count(*) as duplicatecount FROM ecommerce_data group by orderid, date, customerid, product, qu...	0 row(s) returned	0.032 sec / 0.000 sec
✓ 5	11:17:26	select * from ecommerce_data where str_to_date(date, '%Y-%M-%D') is NULL LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.015 sec
✓ 6	11:20:46	select * from ecommerce_data where abs(TotalPrice - (quantity*unitprice)) > 0.01 LIMIT 0, 1000	0 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

Rainy days ahead 33°C

Search

ENG IN 11:21 18-06-2026

```
97
98 -- PRICE VALIDATION
99 • select
100 from ecommerce_data
101 where abs(TotalPrice - (quantity*unitprice)) > 0.01;
102
103
```

Result Grid

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
---------	------	------------	---------	----------	-----------	-----------------	---------------	-------------	----------------	-------------	------------	----------------	------------

fig 4: PRICE VALIDATION

MySQL Workbench - Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: project x project

Limit to 1000 rows

SCHEMAS

Filter objects

▼ decode labs_project1

- Tables
 - dataset for data analytic
 - ecommerce_data
- Views
- Stored Procedures
- Functions
- ▼ parks_and_recreation
 - Tables
 - employee_demographic
 - employee_salary
 - parks_departments
 - Views
 - Stored Procedures
 - Functions
- sys

Administration Schemas

Information

Schema: decode labs_project1

```
30 -- DUPLICATE VALIDATION
31 *
32 count(*) as duplicatecount
33 FROM ecommerce_data
34 group by
35 orderid,
36 date,
37 customerid.
```

Result Grid

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice	duplicatecount
---------	------	------------	---------	----------	-----------	-----------------	---------------	-------------	----------------	-------------	------------	----------------	------------	----------------

Result 30 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	11:03:37	select COUNT(*) AS Total_Records from ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
3	11:06:25	SELECT COUNT(*) AS missingCouponCode From ecommerce_data WHERE couponcode IS NULL or coupon...	1 row(s) returned	0.000 sec / 0.000 sec
4	11:12:35	SELECT *, count(*) as duplicatecount FROM ecommerce_data group by orderid, date, customerid, product, qu...	0 row(s) returned	0.032 sec / 0.000 sec

Object Info Session

33°C Mostly sunny

Search

ENG IN

11:15 18-06-2026

fig 5: DUPLICATE VALIDATION

THANK YOU